Surafel Wondmagegn Techan

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EDUCATION

BSc in Electrical and Computer Engineering 05/2021 – 06/2026
Addis Ababa University Addis Ababa, Ethiopia
CGPA-3.55/4.00(Great Distinction)

UNDERGRADUATE THESIS

Design and Optimization of a Power-Efficient Wearable ECG Patch for Remote Cardiac Telemetry 01/2026 – 07/2026
Addis Ababa, Ethiopia

- Designed and fabricated a compact **4-layer wearable ECG system** for continuous cardiac monitoring and remote telemetry.
- Optimized the power architecture through **power gating and power-domain isolation**, achieving approximately **2–3 days of operation from a 2200 mAh battery**.
- Validated **single-lead ECG acquisition at 250 Hz**, demonstrating reliable **signal capture, storage, and wireless transmission** for remote cardiac monitoring.
- Developed an **ML-based ECG analysis pipeline** for physiological signal processing, providing **automated warnings and estimation of potential cardiac events**.
- Evaluated the complete prototype through **bench testing and end-to-end data transmission**, demonstrating reliable operation from ECG acquisition through remote analysis.

PROFESSIONAL EXPERIENCE

PCB Designer and Embedded Systems Engineer 05/2025 – 07/2026
DAFTech Social ICT Solutions PLC Ethiopia, Addis Ababa

- Designed and developed **custom PCBs for digital water- and energy-metering systems**, including schematic capture, PCB layout, and design documentation.
- Developed **MSP430 firmware and system integration** for metering applications, connecting embedded hardware with backend infrastructure and **LTE telemetry** for remote monitoring.
- Contributed to **power, sensing, communication, and embedded-control optimization** across metering platforms

Embedded System Engineer (Part-Time) 11/2023 – 04/2026
4K Research LABS Addis Ababa, Ethiopia

- Developed embedded systems from **prototyping through testing**, including robotics and low-power automation.
- Integrated **sensors, microcontrollers, and firmware** while optimizing system performance and reliability.
- Mentored team members in **embedded systems development and prototyping**.

Teaching Assistant 07/2024 – 09/2024
4K Research LABS Addis Ababa, Ethiopia

- Assisted in teaching Data Analysis Fundamentals
- Assisted in teaching Arduino Mastery courses.

LANGUAGES

Amharic : Native English: C1

SKILLS

Embedded Systems
Using ATtiny , Raspberry Pi, Arduino, ESP32,STM32, MSP430

PCB & Schematic Design
Altium, KiCad, EasyEDA, Proteus (schematic capture, layout, PCBA)

Teaching & Mentoring
Course Assistance, Student Mentorship, Lab Support

Robotics
Autonomous Navigation, Path Planning

System Optimization
Energy-efficient Design

PROGRAMMING LANGUAGE

- C++
- Python
- Java
- Assembly Language
- VHDL
- C

CERTIFICATES

- Data Analysis Fundamentals — 4K LABS
- Low-Level Computing for Computer Science & Engineering, 2025
- Certificate of Recognition for Excellence in Assistant Teaching, 4k Labs, 2024
- Kifiya Al Mastery Training Program -Kifiya Financial Technology PLC,2025